

Sofija Dimitrijevic

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EDUCATION

California Polytechnic State University, San Luis Obispo Sep. 2022 – Jun. 2027
B.S./M.S. (Blended) in Computer Science with AI/ML Concentration, Minor in Linguistics GPA: 3.7

- Dean's List & President's Honor List (Fall 2022–Fall 2024, Spring 2025); SWE Scholarship Recipient 2025
- Relevant Coursework:** Speech and Language Processing, Machine Learning, Operating Systems, Software Engineering, Database Systems, Distributed & Parallel Computing, Computer Networks, Applied Regression Analysis, Data Structures & Algorithms, Object-Oriented Programming in Java, Systems Programming

TECHNICAL SKILLS

Languages: Python, Java, C/C++, C#, SQL, MATLAB, JavaScript, TypeScript, HTML/CSS, Racket, R
Frameworks/Tools: PyTorch, TensorFlow, scikit-learn, NumPy, Pandas, Matplotlib, Optuna, SciPy, Hugging Face, CUDA, .NET, WPF, AsyncIO, MCP, Node.js, Express, MongoDB, Git, Docker, Figma, Jupyter, Google Colab

EXPERIENCE

Software Engineering Intern Jun. 2025 – Sep. 2025
HP Inc. *Houston, TX*

- Designed and deployed an **AI-powered telemetry service** with a Random Forest model, **boosting prediction accuracy by 50%** and reducing hardware validation costs.
- Built a C#/.NET desktop client with IPC messaging for **real-time inference at scale**, enabling deployment across thousands of devices.
- Integrated CI/CD pipelines and automated benchmarking, cutting validation time by **30%** and improving reproducibility across teams.

Physics Student Researcher Jul. 2024 – Sep. 2024
Cal Poly Partners *San Luis Obispo, CA*

- Extended the Active Finite Voronoi (AFV) model by integrating flocking behavior to study cancer progression and wound healing.
- Improved simulation runtime by **10%** through vectorization and translated legacy MATLAB code to Python, enabling broader use and scalability in HPC environments.

PROJECTS

MCP Server–Client Protocol | *Python, AsyncIO, Pydantic* Aug. 2025

- Implemented asynchronous server–client system using Anthropic's MCP framework with support for custom tools and resource APIs.
- Optimized concurrency for tool calls and resource reads, **improving throughput by 40%** and reducing latency for multi-user sessions.

Gather: Inventory Management Web App | *React, TypeScript, REST API* Apr. 2024 – Jun. 2024

- Built full-stack inventory management system with REST APIs and CRUD features in a 5-person Agile team.
- Refactored backend into modular services, reducing code duplication by **50%** and supporting CI testing pipelines.

Moment: Burnout Prediction Web App (Hackathon) | *TensorFlow, scikit-learn* Jan. 2024

- Developed ML models for predicting hospital staff burnout with 80% accuracy; collaborated on web app prototype.
- Contributed in a team of 5 to deliver prototype in 24 hours; won **Best Design** award out of 15 teams.

LEADERSHIP & CAMPUS INVOLVEMENT

Women Involved in Software and Hardware (WISH) Sep. 2022 – Present
Vice President (Current), Technical Director (Previous) *San Luis Obispo, CA*

- Overseeing strategy and operations for 300+ member organization; managing cross-functional officer board of 20.
- Led team to deliver 20+ weekend technical workshops and build days annually, fostering **technical collaboration** and onboarding new members into software engineering practices.

Society of Women Engineers (SWE) Sep. 2022 – Present
Scholarship Recipient, Active Member *San Luis Obispo, CA*